Two-Minute Interview Narration

A concise, technically accurate explanation of the Customer Segmentation project.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

Narration

“I worked on a retail customer-segmentation project using the SmartCart customer dataset. It contains 2,240 customer records and 22 columns covering demographics, household composition, income, purchase recency, product spending, purchase channels and campaign response. The objective was to group customers with similar profiles so the business could design more relevant marketing strategies.”

“I first handled 24 missing Income values using median imputation. I engineered Age, customer tenure, Total_Spending and Total_Children, grouped education into three categories and simplified marital status into living arrangement. I removed identifier and redundant source columns, applied basic age and income filters, and one-hot encoded Education and Living_With. That left 2,236 records and 18 numeric features.”

“Since the features had different units, I applied StandardScaler before dimensionality reduction. I used PCA to reduce the 18-dimensional matrix to three components; those components explained about 44.95 percent of the variance. I explored K-Means for values of K from 1 to 10 using the elbow method. KneeLocator suggested four clusters, and I also plotted silhouette scores as a second diagnostic.”

“An important implementation detail is that the notebook uses K-Means to explore K and visualise clusters, but it uses Ward AgglomerativeClustering with four clusters for the final customer labels and cluster-profile table. The profiles show two broad spending tiers: clusters 0 and 2 have lower income and spending, while clusters 1 and 3 have higher income and spending. Household composition and channel activity further distinguish the groups.”

“There are limitations. The notebook fits preprocessing on the full dataset and does not perform stability or holdout validation. It also includes Response as a clustering feature, so the higher response rate in cluster 3 is not independent evidence of future campaign performance. For a prospective marketing use case, I would exclude Response, build a reusable preprocessing pipeline, test cluster stability and validate segment-specific offers through randomized experiments.”

Likely interview questions

1. Why is this unsupervised learning? There is no ground-truth segment label; the algorithm discovers groups from feature similarity.

2. Why standardize? Distance-based algorithms can be dominated by variables with larger numeric scales.

3. Why PCA? It compresses correlated features and can make clustering/visualisation more manageable; here three components retain 44.95% of variance.

4. How did you choose K? K-Means inertia and KneeLocator suggested K=4; silhouette scores were plotted as an additional diagnostic, but their numeric values were not printed.

5. Which algorithm generated final labels? Ward AgglomerativeClustering with four clusters. K-Means was used for K exploration and visualisation.

6. What is Ward linkage? It merges clusters to minimise the increase in within-cluster sum of squares.

7. Why is Response a concern? If it is a past campaign outcome, including it can leak outcome information into a segmentation intended to target future campaigns.

8. Why not accuracy? There are no known correct cluster labels against which to calculate classification accuracy.

9. What are the four segments? Two lower-spend and two higher-spend groups, with living arrangement and channel differences; IDs are arbitrary.

10. What would you improve? Exclude Response for prospective segmentation, justify outlier filters, fit preprocessing on a reference cohort, assess stability and test marketing actions experimentally.

11. Can new customers be assigned to Agglomerative clusters? Not directly with scikit-learn's standard AgglomerativeClustering; define a principled assignment approach or use a centroid-based model such as K-Means.

12. Does a high response rate prove campaign effectiveness? No. It is descriptive, and Response was included as an input feature; randomized experiments are needed for causal lift.